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ELECTRONIC APPARATUS AND CHASSIS MEMBER

Abstract

An electronic apparatus includes a chassis member having standing walls on edge portions of the chassis member, a display panel having a display surface and a rear surface opposite to the display surface supported by an inner surface of the chassis member, and a glass plate configured to cover the display surface of the display panel, in which an edge surface of an outer edge portion protruding from an outer peripheral side surface of the display panel faces the standing wall, in which an outer edge portion of the glass plate has a corner portion, the standing wall has a corner wall portion that is disposed around the corner portion and has an interior wall surface facing the corner portion, and the interior wall surface of the corner wall portion is provided with a recessed hole recessed in a direction away from the corner portion.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electronic apparatus and a chassis member.

Description of the Related Art

[0002] An electronic apparatus such as a laptop PC includes a display panel configured with a liquid crystal display or the like. For example, Japanese Unexamined Patent Application Publication No. 2023-005541 discloses a configuration in which a display panel is supported by the inner surface of a chassis member having standing walls on the edge portions thereof.

SUMMARY OF THE INVENTION

[0003] In the configuration of Japanese Unexamined Patent Application Publication No. 2023-005541, a display surface of a display panel is covered with a glass plate. The glass plate has a corner portion. The standing wall of the chassis member that supports the display panel also has corner wall portions that are disposed to surround the corner portion of the glass plate. Therefore, when such an electronic apparatus receives an impact from being dropped or the like, there is a concern that the standing wall may be deformed by the impact and collide with the corner portion of the glass plate, possibly causing breakage or damage to the glass.

[0004] In this regard, in the configuration of Japanese Unexamined Patent Application Publication No. 2023-005541, a thin plate piece provided in a bezel member made of a resin is interposed between the edge surface of the glass plate and the standing wall. Therefore, breakage or the like of the glass due to an impact from being dropped or the like mentioned above, is suppressed to some extent.

[0005] Incidentally, it is desired that the bezel width surrounding the outer circumference of the display panel be configured to be as narrow as possible to improve the appearance quality. In the configuration of Japanese Unexamined Patent Application Publication No. 2023-005541, the bezel width is increased by the thickness of the bezel member interposed between the glass plate and the standing wall. On the other hand, in a case in which the bezel member is eliminated from this configuration, there is a concern that the glass may directly interfere with the standing wall and be broken or damaged.

[0006] In particular, in the electronic apparatus, there is a model that can be used as a tablet PC by rotating the two chassis up to 360 degrees (see, for example, Japanese Unexamined Patent Application Publication No. 2017-91150). This type of electronic apparatus may be used at a higher position than a general clamshell type PC, for example, above the waist of a standing user. Therefore, in this configuration, the impact at the time of being dropped is further increased, and the breakage of the glass or the like is further problematic.

[0007] The present invention has been made in consideration of the above-described problems of the related art, and an object of the present invention is to provide an electronic apparatus and a chassis member capable of suppressing a glass plate from being broken or damaged.

[0008] An electronic apparatus according to a first aspect of the present invention includes a chassis member having standing walls on edge portions of the chassis member, a display panel having a display surface and a rear surface opposite to the display surface supported by an inner surface of the chassis member, and a glass plate configured to cover the display surface of the display panel, in which an edge surface of an outer edge portion protruding from an outer peripheral side surface of the display panel faces the standing wall, in which an outer edge portion of the glass plate has a corner portion, the standing wall has a corner wall portion that is disposed

around the corner portion and has an interior wall surface facing the corner portion, and the interior wall surface of the corner wall portion is provided with a recessed hole recessed in a direction away from the corner portion.

[0009] A chassis member according to a second aspect of the present invention is a chassis member for supporting a display assembly including a display panel having a display surface and a glass plate that covers the display surface and is provided such that an outer edge portion of the glass plate protrudes from an outer peripheral side surface of the display panel, the chassis member including a plate portion configured to support a rear surface of the display panel on a side opposite to the display surface, and a standing wall configured to stand upright from an edge portion of the plate portion, in which the standing wall has a corner wall portion with a recessed hole provided on an interior wall surface of the standing wall.

[0010] According to the above-described aspect of the present invention, it is possible to suppress a glass plate from being broken or damaged.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a schematic plan view of an electronic apparatus according to an embodiment as viewed from above.

[0012] FIG. 2 is a schematic front view of a chassis member.

[0013] FIG. 3 is a rear view of a display assembly.

[0014] FIG. 4 is an enlarged perspective view of a part of the chassis member.

[0015] FIG. 5A is an enlarged plan view of a corner wall portion of a first chassis and a peripheral portion thereof.

[0016] FIG. 5B is a plan view showing a state in which the display assembly is detached from the chassis member shown in FIG. 5A.

[0017] FIG. 6 is a view showing a shape of the chassis member shown in FIG. 5B.

[0018] FIG. 7 is a schematic cross-sectional view along a line VII-VII in FIG. 5A.

DETAILED DESCRIPTION OF THE INVENTION

[0019] Hereinafter, suitable embodiments of the electronic apparatus and the chassis member according to the present invention will be described in detail with reference to the accompanying drawings.

[0020] FIG. 1 is a schematic plan view of an electronic apparatus **10** according to an embodiment as viewed from above. The electronic apparatus **10** of the present embodiment has a configuration in which a first chassis **11** and a second chassis **12** are connected to each other by a hinge **14** to be rotationally movable relative to each other.

[0021] The electronic apparatus **10** of the present embodiment is a so-called convertible type PC that can be used as a laptop (laptop mode) or a tablet type PC (tablet mode) depending on an angular position between the chassis **11** and **12**. The electronic apparatus **10** can set the chassis **11** and **12** at a desired angular position within an angular range of 0 degrees to 360 degrees. The 0 degrees is a position at which the surface directions of the chassis **11** and **12** are parallel to each other and a front surface **11a** of the first chassis **11** and the surface (keyboard **16**) of the second chassis **12** face each other. The 360 degrees is a position at which the surface directions of the chassis **11** and **12** are parallel to each other and a rear surface **11b** of the first chassis **11** and the bottom surface (surface opposite to the keyboard **16** side) of the second chassis **12** face each other. The electronic apparatus **10** may be various electronic computers such as a general-purpose laptop PC or a tablet type PC having a single plate shape, which can be set at a desired angular position within an angular range of approximately 0 degrees to 180 degrees between the chassis **11** and **12**, in addition to the convertible type PC.

[0022] The second chassis **12** is a flat box having a rectangular outer shape in a plan view, and is adjacent to the first chassis **11**. Inside the second chassis **12**, various electronic components such as a motherboard on which a CPU and the like are mounted, a battery device, a memory, an antenna device are accommodated. The keyboard **16** and a touch pad **17** face the upper surface of the second chassis **12**.

[0023] The first chassis **11** has a rectangular outer shape in a plan view and is a flat box that is thinner than the second chassis **12**. A display panel **18** is mounted on the first chassis **11**.

[0024] Hereinafter, the first chassis **11** and each of the components mounted thereon will be described based on the direction as seen by a user viewing a display surface **18a** of the display panel **18**, with the left-right direction respectively referred to as X1 and X2 directions, the up-down direction respectively referred to as Y1 and Y2 directions, and the depth direction respectively referred to as Z1 and Z2 directions. The X1 and X2 directions are sometimes collectively referred to as the X direction, and the Y1 and Y2 directions and the Z1 and Z2 directions are sometimes similarly referred to as the Y direction and the Z direction.

[0025] The display surface **18a** of the display panel **18** faces the Z1-side surface (front surface **11a**) of the first chassis **11**. The display surface **18a** is the surface on which the video or image is displayed. The display panel **18** is configured with, for example, a liquid crystal display or an organic EL display. The display panel **18** has a structure in which, for example, glass, a liquid crystal layer, a light guide plate, or the like are laminated, and the outer peripheral edge portions of each layer are fixed to each other with a double-sided tape, an adhesive, or the like.

[0026] The display surface **18a** of the display panel **18** is covered with a glass plate **19**. The glass plate **19** is a cover glass that covers substantially the entire front surface **11a** of the first chassis **11** including the display panel **18**. The glass plate **19** can be configured with touch glass corresponding to a touch operation. The glass plate **19** may not support a touch operation.

[0027] The first chassis **11** has a chassis member **20**. FIG. 2 is a schematic front view of the chassis member **20** constituting the first chassis **11**. FIG. 2 shows an inner surface **20a** (back surface of the rear surface **11b**) of the chassis member **20** and predetermined components installed on the inner surface **20a**.

[0028] The chassis member **20** has a rectangular plate portion **21** and a standing wall **22** standing upright from the outer peripheral edge portion of the plate portion **21**. The chassis member **20** has a shallow bathtub shape. The plate portion **21** forms the Z2-side surface (rear surface **11b**) of the first chassis **11**. The standing wall **22** is provided to surround most of the four sides of the plate portion **21** and stands in the Z1 direction. The chassis member **20** is made of, for example, metal and is formed of, for example, an aluminum alloy or a magnesium alloy. The chassis member **20** may be made of a resin. The chassis member **20** may be configured by joining a frame material that constitutes the standing wall **22** to the periphery of the plate portion **21** formed of, for example, a carbon fiber reinforced resin.

[0029] The chassis member **20** supports a rear surface **18b** of the display panel **18** on a side opposite to the display surface **18a** with the inner surface **20a**. Specifically, the rear surface **18b** of the display panel **18** is fixed to the inner surface **20a** of the plate portion **21** by using a double-sided adhesive tape **24**.

[0030] The hinge **14** is connected to the Y2-side edge portion (edge portion **11c**) of the first chassis **11** in a pair of left and right directions. A reference numeral **14a** in FIG. 2 denotes a hinge fixing portion to which the hinge **14** is fixed. In a case in which the chassis member **20** is made of metal, the hinge fixing portion **14a** can form a screw thread by direct processing on the plate portion **21**. In a case where the chassis member **20** is made of a resin, the hinge fixing portion **14a** can be formed of a metal bracket to which the hinge **14** can be screwed. The chassis member **20** has a cutout portion **20b** in which the hinge **14** is disposed on a side portion of the hinge fixing portion **14a**.

[0031] The display panel **18** according to the present embodiment can be used as a display

assembly **26** in which the glass plate **19** is fixed to the display surface **18a**. As described above, the rear surface **18b** of the display panel **18** is fixed to the chassis member **20** by the double-sided adhesive tape **24**. In this case, the outer periphery of the glass plate **19** extends along the inner periphery of the standing wall **22**. As a result, the display assembly **26** is attached to the chassis member **20**.

[0032] Next, an assembly structure of the display assembly **26** with respect to the chassis member **20** will be described.

[0033] FIG. **3** is a rear view of the display assembly **26**. FIG. **4** is an enlarged perspective view of a part of the chassis member **20**, specifically, a corner wall portion CW**1** and a peripheral portion thereof. FIG. **5A** is an enlarged plan view of the corner wall portion CW**1** of the first chassis **11** and a peripheral portion thereof. FIG. **5B** is a plan view in which the display assembly **26** is detached from the chassis member **20** shown in FIG. **5A**. FIG. **6** is a view showing a shape of the chassis member **20** shown in FIG. **5B**. FIG. **7** is a schematic cross-sectional view taken along the line VII-VII in FIG. **5A**.

[0034] First, a schematic configuration of the standing wall **22** will be described.

[0035] As shown in FIGS. **2** and **4** to **7**, the standing wall **22** has an interior wall surface **22a** facing the inside of the first chassis **11** and an exterior wall surface **22b** facing the outside of the first chassis **11** on a side opposite to the interior wall surface **22a**. The interior wall surface **22a** faces a direction of an outer peripheral side surface **18c** of the display panel **18**. The exterior wall surface **22b** is painted on the surface as necessary, and forms an outer surface of the chassis member **20**.

[0036] The standing wall **22** extends to surround the entire periphery of the first chassis **11**. The standing wall **22** has a pair of horizontal walls **30** and **31** extending in the X direction and a pair of vertical walls **32** and **33** extending in the Y direction. The walls **30** to **33** are linear standing walls (linear wall portions) extending along the X direction or the Y direction.

[0037] One horizontal wall **30** constitutes the Y1-side edge portion (edge portion **11d**) of the first chassis **11**. The horizontal wall **30** extends over substantially the entire length of the edge portion **11d**. The first chassis **11** of the present embodiment has a bulging portion **34** that bulges outward (Y1 side) at a central portion of the edge portion **11d** in the longitudinal direction (see FIG. **2**). The bulging portion **34** expands a space between the standing wall **22** (horizontal wall **30**) and the outer peripheral side surface **18c** of the display panel **18**, and secures an installation space for a high-performance camera module **36**. The camera module **36** includes a camera lens **36a**. The camera lens **36a** is a camera that can be used for video capturing or the like, has a lens for focusing in the center, and has an image sensor provided on a back side of the lens. The camera module **36** can further include an IRLED (infrared camera), an illuminance sensor, and the like.

[0038] The vertical walls **32** and **33** extend along the Y direction and are perpendicular to the horizontal walls **30** and **31**, respectively. As a result, corner wall portions CW**1** to CW**4** are formed at the four corners of the standing wall **22**. The horizontal wall **30** and the vertical walls **32** and **33**, which are linear wall portions, are respectively continuous from both ends of the corner wall portions CW**1** and CW**2** and extend in directions perpendicular to each other.

[0039] As shown in FIGS. **2** and **4**, the horizontal wall **30** can have a support surface **22c** below the interior wall surface **22a** (on the base side of the standing wall **22**). The support surface **22c** is an upper surface of a high ridge that is one step higher than the inner surface **20a**, and protrudes to the Y1 side from a lower end of the interior wall surface **22a**. The support surface **22c** can support resin members **52** described later. The support surface **22c** is provided on at least the horizontal wall **30** on the Y1 side, but may be similarly provided on the other walls **31** to **33**.

[0040] The horizontal wall **30** of the present embodiment has a recess **24a** on the side of the corner wall portions CW**1** and CW**2**. The recess **24a** is formed by cutting out a part of the support surface **22c** in the Y1 direction. The recess **24a** can be used to dispose the end portions of the double-sided adhesive tape **24**. In the horizontal wall **30**, a recess **37** in which the support surface **22c** is cut out in an arc shape in the Y1 direction is further formed between the recess **24a** and the corner wall

portions CW1 and CW2. The recess 37 is an escape portion for preventing the corners of the display panel 18 from colliding with the standing wall 22 due to an impact when the electronic apparatus 10 is dropped on a floor surface or the like.

[0041] The corner wall portions CW1 and CW2 of the standing wall 22 can have all or a part of recessed holes 40, an inclined portion 42, and a thick-walled portion 44. Details of the recessed holes 40 and the like will be described later.

[0042] Next, a configuration of the display assembly 26 will be described.

[0043] As shown in FIGS. 3, 5A, and 7, the display assembly 26 is a laminate in which the glass plate 19 is fixed to the display surface 18a of the display panel 18.

[0044] The glass plate 19 can be fixed by an adhesive member 50 that is transparent to the display surface 18a. The surface area of the glass plate 19 is slightly larger than the surface area of the display panel 18. Therefore, the glass plate 19 has an eave-like outer edge portion 19a that protrudes outward from the outer peripheral side surface 18c of the display panel 18.

[0045] In the glass plate 19, an edge surface 19b of the outer edge portion 19a faces the interior wall surface 22a of the standing wall 22 with a predetermined gap C therebetween. The gap C is, for example, 0.2 mm. Specifically, among the four standing walls 22, the horizontal wall 30 on the Y1 side and the left and right vertical walls 32 and 33 have the interior wall surface 22a facing the edge surface 19b with the gap C therebetween. Since the hinge 14 or the components related to the wiring line are disposed between the edge surface 19b and the horizontal wall 31 on the Y2 side, the gap between the interior wall surface 22a and the edge surface 19b is wide, for example, 10 mm or more.

[0046] Corner portions C1 to C4 that are curved along the corner wall portions CW1 to CW4 of the chassis member 20 are formed at four corners of the glass plate 19. The corner portions C1 and C2 provided at one end and the other end of the outer edge portion 19a on the Y1 side in the longitudinal direction face the interior wall surfaces 22a of the corner wall portions CW1 and CW2 with the gap C therebetween. The corner portions C3 and C4 provided at both ends of the outer edge portion 19a on the Y2 side are disposed with the same gap as the wide gap (for example, 10 mm or more) between corner wall portions CW3 and CW4 and the horizontal wall 31 described above.

[0047] The display assembly 26 can further have a pair of left and right resin members 52 and 52. The resin member 52 can be fixed to a back surface 19c of the outer edge portion 19a of the glass plate 19 on the Y1 side by using an adhesive member 54 (see FIG. 7). The adhesive member 54 can be exemplified by a double-sided adhesive tape, an adhesive, or a bonding agent. Hereinafter, the outer edge portion 19a of the glass plate 19 that extends in the X direction on the Y1 side may be referred to as an “outer edge portion 19a1”. A bulging portion 19a2 corresponding to the shape of the bulging portion 34 is formed on the outer edge portion 19a1 on the Y1 side at a central portion in the longitudinal direction. The chassis member 20 may be configured not to have the bulging portion 34. In this case, the bulging portion 19a2 of the glass plate 19 can also be omitted, and the outer edge portion 19a1 can be formed in a linear shape extending in the X direction.

[0048] As shown in FIGS. 3 and 7, the resin member 52 is a strip-shaped resin part extending in the X direction. The resin member 52 on the X1 side extends from a position close to the corner portion C1 to the front of the bulging portion 19a2. The resin member 52 on the X2 side extends from a position close to the corner portion C2 to the front of the bulging portion 19a2. The resin member 52 is disposed on the support surface 22c of the horizontal wall 30 with a slight gap. Each resin member 52 can form a protrusion or a hook that is aligned with or engages with, for example, a recessed portion or a plate piece formed on the support surface 22c of the horizontal wall 30.

[0049] In the glass plate 19, the corner portions C1 and C2 at both ends of the outer edge portion 19a1 face the corner wall portions CW1 and CW2 of the standing wall 22 with the narrow gap C therebetween. That is, the corner wall portions CW1 and CW2 are respectively disposed to be close to each other so as to surround the corner portions C1 and C2. In this case, a resin part such as a

bezel member is not interposed in the gap C. As a result, the electronic apparatus **10** reduces the width of the bezel surrounding the display assembly **26**.

[0050] On the other hand, the corner portions **C3** and **C4** on the Y2 side are disposed at positions separated from the corner wall portions **CW3** and **CW4** by a certain distance. This is because the metal bracket **14a** or the like is disposed on the edge portion **11c**, and thus there is a limit to the reduction in the width of the bezel. Depending on the structure of the hinge **14**, the metal bracket **14a**, or the like, the corner wall portions **CW3** and **CW4** may be disposed close to each other so as to surround the corner portions **C3** and **C4**.

[0051] As described above, in the electronic apparatus **10** of the present embodiment, at least the corner portions **C1** and **C2** and the corner wall portions **CW1** and **CW2** face each other at positions close to each other, for example, with the gap C of 0.2 mm therebetween. Therefore, when the electronic apparatus **10** receives an impact from being dropped with the corner wall portions **CW1** and **CW2** facing down, particularly in a manner known as corner drop, there is a concern that the corner wall portions **CW1** and **CW2** of the standing wall **22** will be deformed toward the inside of the first chassis **11**. In this case, there is a concern that the deformed corner wall portions **CW1** and **CW2** collide with the edge surface **19b** and the glass plate **19** is broken or damaged. In particular, the chassis member **20** can be made of a metal such as an aluminum alloy. In this case, since the metal standing wall **22** abuts against the glass plate **19**, there is a concern that the glass plate **19** may be broken or damaged.

[0052] Therefore, as shown in FIGS. **4**, **5B**, and **7**, the electronic apparatus **10** of the present embodiment can be provided with the recessed holes **40**, the inclined portion **42**, and the thick-walled portion **44** in the corner wall portions **CW1** and **CW2**.

[0053] The recessed holes **40**, the inclined portion **42**, and the thick-walled portion **44** have a structure for suppressing the corner wall portions **CW1** and **CW2** from colliding with the glass plate **19**. It is most preferable that the recessed holes **40**, the inclined portion **42**, and the thick-walled portion **44** are provided at the same time. Only one or two of the recessed holes **40**, the inclined portion **42**, and the thick-walled portion **44** may be provided, for example, only the recessed holes **40** may be provided. In a case in which the corner portions **C3** and **C4** and the corner wall portions **CW3** and **CW4** are close to each other or the like, the recessed holes **40** and the like can also be provided in the corner wall portions **CW3** and **CW4**.

[0054] Hereinafter, as shown in FIGS. **4** to **7**, the configuration of the corner wall portion **CW1** on the X1 side will be typically described. The corner wall portion **CW2** on the X2 side can have the same or similar structure as the corner wall portion **CW1** shown in FIGS. **4** to **7**, except that the corner wall portion **CW2** has a left-right symmetrical structure with the corner wall portion **CW1**. Therefore, the detailed description of the corner wall portion **CW2** will be omitted.

[0055] First, the recessed hole **40** is a hollowed-out recess provided in the interior wall surface **22a** of the corner wall portion **CW1**. The recessed hole **40** is recessed in a direction away from the corner portion **C1** of the glass plate **19**, that is, in a direction from the interior wall surface **22a** toward the exterior wall surface **22b**. The recessed hole **40** is curved in a bow shape along the outer edge of the corner portion **C1**. The recessed hole **40** has a shape that allows the interior wall surface **22a** to be prevented from colliding with the corner portion **C1** when the corner wall portion **CW1** is deformed and moved toward the glass plate **19** side.

[0056] As shown in FIG. **7**, in a case in which the standing direction (Z1 direction) of the standing wall **22** is used as a reference, an upper end **40a** of the recessed hole **40** can be at a height position equal to or lower than the surface **19d** of the glass plate **19**. The surface **19d** of the glass plate **19** is a surface opposite to the back surface **19c** and is a surface that constitutes most of the front surface **11a** of the first chassis **11**. In the configuration example shown in FIG. **7**, the upper end **40a** is at a position (Z2 side) lower than the surface **19d** by a height H. For example, the height H can be set to be approximately equal to the height of a chamfered portion **19d1** provided on the edge surface **19b** of the glass plate **19** in the Z direction. The height H of the present embodiment is, for example, 0.1

mm.

[0057] In a case in which the standing direction of the standing wall **22** is used as a reference, the lower end **40b** of the recessed hole **40** can be at a position (Z2 side) lower than the back surface **19c** of the glass plate **19**. In the configuration example shown in FIG. 7, the Z-direction height of the back surface **19c** is near the center of the Z-direction height of the recessed hole **40**.

[0058] Next, the inclined portion **42** is an inclined surface provided on the interior wall surface **22a** of the corner wall portion CW1. The inclined portion **42** is gradually inclined to the display panel **18** side (plate portion **21** side) toward the base side (Z2 side) of the corner wall portion CW1. The inclined portion **42** is curved along the interior wall surface **22a** that is curved along the corner wall portion CW1. The inclined portion **42** is a gradient provided at a base portion of the corner wall portion CW1. The inclined portion **42** has a bank shape that wraps around the inner periphery of the corner wall portion CW1. The inclined portion **42** is a reinforcing shape for suppressing the deformation of the corner wall portion CW1 toward the glass plate **19** side when the corner drop of the electronic apparatus **10** happens.

[0059] The length of the inclined portion **42** can extend to the horizontal wall **30** and the vertical wall **32** adjacent to the corner wall portion CW1. In the inclined portion **42** of the configuration example shown in FIGS. 4 and 5A, one end portion **42a** reaches the interior wall surface **22a** of the horizontal wall **30**, and the other end portion **42b** reaches the interior wall surface **22a** of the vertical wall **32**.

[0060] The corner wall portion CW1 has high rigidity by including the inclined portion **42**. In particular, the corner wall portion CW1 can ensure sufficient strength by the inclined portion **42** while the recessed hole **40** is provided. As shown in FIGS. 4 and 7, the recessed hole **40** is formed so as to hollow out a part of the inclined portion **42**. In other words, the upper end of the inclined portion **42** can be set to be above (Z1 side) the lower end **40b** of the recessed hole **40**.

[0061] Next, the thick-walled portion **44** has a plate thickness t_1 of the corner wall portion CW1 larger than a plate thickness t_2 of the horizontal wall **30** and vertical wall **32** that are adjacent to each other (see FIG. 5B). That is, the corner wall portion CW1 can have a larger wall thickness than the linear wall portions (walls **30**, **32**, and **33**) extending along the outer peripheral side surface **18c**. The thick-walled portion **44** is a reinforcing shape for increasing the plate thickness of the corner wall portion CW1 itself to suppress deformation of the corner wall portion CW1 toward the glass plate **19** when the corner drop of the electronic apparatus **10** happens.

[0062] As shown in FIG. 5B, the interior wall surface **22a** and the exterior wall surface **22b** of the corner wall portion CW1 each have an arc shape (quarter circle shape). The corner wall portion CW1 has a gradually increasing plate thickness from boundaries B1 and B2 with the linear wall portions (walls **30** and **32**) toward a middle M of the corner. As a result, the plate thickness of the corner wall portion CW1 is changed such that the plate thickness is maximized in the vicinity of the middle M. The boundary B1 indicates a boundary on the interior wall surface **22a** side. The boundary B2 indicates a boundary on the exterior wall surface **22b** side.

[0063] It is preferable that the thick-walled portion **44** has a uniform symmetrical structure in which a portion extending to the horizontal wall **30** side and a portion extending to the vertical wall **32** side are centered on the middle M of the corner wall portion CW1. As a result, it is possible to suppress occurrence of an unintended stress concentration at various places of the thick-walled portion **44** during the corner drop and to more effectively suppress deformation of the corner wall portion CW1.

[0064] A virtual circle **22a1** shown by a one-dot chain line in FIG. 6 is obtained by extending a locus of a circle passing through the interior wall surface **22a** of the corner wall portion CW1. A virtual circle **22b1** is obtained by extending a locus of a circle passing through the exterior wall surface **22b** of the corner wall portion CW1. A virtual circle **40c** is obtained by extending a locus of a circle passing through the bottom portion of the recessed hole **40**. A center point O1 in FIG. 6 indicates a center of the virtual circle **22a1**. A center point O2 indicates the center of the virtual

circle **22b1**, **40c**. Radii $r1$, $r2$, and $r3$ in FIG. **6** indicate radii of the virtual circles **22a1**, **22b1**, and **40c**, respectively.

[0065] As shown in FIG. **6**, the center **O1** of the virtual circle **22a1** drawing the interior wall surface **22a** is at a position offset to the inside of the chassis member **20** from the center **O2** of the virtual circle **22b1** drawing the exterior wall surface **22b**. More specifically, the centers **O1** and **O2** are on a 45-degree line passing through the middle **M** of the corner wall portion **CW1**, and the center **O2** is at a position closer to the interior wall surface **22a** than the center **O1**. In the present embodiment, the radius $r1$ of the virtual circle **22a1** can be slightly smaller than or the same as the radius $r2$ of the virtual circle **22b1**. The radius $r3$ of the virtual circle **40c** can be set to be the same as a radius of the corner portion **C1** of the glass plate **19**.

[0066] Accordingly, in the corner wall portion **CW1**, the boundary **B1** between the interior wall surface **22a** and the linear wall portions (walls **30** and **32**) and the boundary **B2** between the exterior wall surface **22b** and the linear wall portions (walls **30** and **32**) are misaligned in the X direction or the Y direction (see FIG. **6**). As a result, the corner wall portion **CW1** can form the thick-walled portion **44** whose thickness increases uniformly from the boundaries **B1** and **B2** with the walls **30** and **32** toward the middle **M**.

[0067] As described above, the electronic apparatus **10** of the present embodiment can include the chassis member **20** having the standing wall **22** on the edge portions thereof, the display panel **18** having the rear surface **18b** supported by the inner surface **20a** of the chassis member **20**, and the glass plate **19** that covers the display surface **18a** of the display panel **18** and of which the edge surface **19b** of the outer edge portion **19a** protruding from the outer peripheral side surface **18c** faces the standing wall **22**. The standing wall **22** has the corner wall portions **CW1** and **CW2** disposed to surround the corner portions **C1** and **C2** of the glass plate **19** and having the interior wall surfaces **22a** facing the corner portions **C1** and **C2**. The interior wall surfaces **22a** of the corner wall portions **CW1** and **CW2** are provided with recessed holes **40** recessed in a direction away from the corner portions **C1** and **C2**.

[0068] The chassis member **20** of the present embodiment can be used to support the display assembly **26** including the display panel **18** and the glass plate **19** provided such that the outer edge portion **19a** protrudes from the outer peripheral side surface **18c**. The chassis member **20** includes the plate portion **21** for supporting the rear surface **18b** of the display panel **18** and a standing wall **22** that stands upright from an edge portion of the plate portion **21**. The standing wall **22** has the corner wall portions **CW1** and **CW2** in which the interior wall surface **22a** is provided with the recessed holes **40**.

[0069] In such electronic apparatus **10** and chassis member **20**, the corner wall portions **CW1** and **CW2** may be deformed toward the inside of the first chassis **11** due to an external force such as corner drop. Even in this case, the electronic apparatus **10** can allow the edge surface **19b** of the glass plate **19** to escape with the recessed hole **40** provided on the interior wall surfaces **22a** of the corner wall portions **CW1** and **CW2**. As a result, the electronic apparatus **10** can suppress the standing wall **22** from colliding with the edge surface **19b** of the glass plate **19**. Accordingly, the electronic apparatus **10** can suppress the glass plate **19** from being broken or damaged due to an impact from being dropped or the like.

[0070] Here, as shown in FIG. **7**, a case in which an external force **F** is applied to the corner wall portion **CW1** (**CW2**) is considered. In this case, it is assumed that the standing wall **22** is deformed in a direction (movement direction) **D** indicated by an arrow in the drawing. This is because the rigidity of the upper portion (the distal end portion on the **Z1** side) of the standing wall **22** in the standing direction is the lowest. Then, the standing wall **22** is moved to the glass plate **19** side such that the recessed hole **40** engulfs the edge surface **19b**. As a result, the interior wall surface **22a** is prevented from colliding with the edge surface **19b**.

[0071] Therefore, when the upper end **40a** of the recessed hole **40** is at a height position equal to or lower than the surface **19d** of the glass plate **19** in the **Z** direction, interference with the edge

surface **19b** can be effectively prevented. In addition, when the lower end **40b** of the recessed hole **40** is at a position lower than the back surface **19c** of the glass plate **19** in the Z direction, interference with the edge surface **19b** can be effectively prevented.

[0072] In particular, the electronic apparatus **10** may be used as a tablet type PC by rotating the chassis **11** and **12** by 360 degrees. In this case, the electronic apparatus **10** may be used at a position higher than a general clamshell type PC, for example, above the waist of the standing user. However, even in a case in which the electronic apparatus **10** is dropped from such a height, the recessed hole **40** can suppress the glass plate **19** from being broken or the like.

[0073] The electronic apparatus **10** and the chassis member **20** can also have the inclined portion **42** on the interior wall surfaces **22a** of the corner wall portions CW1 and CW2. The electronic apparatus **10** and the chassis member **20** can also have the thick-walled portion **44** in which the plate thickness **t1** of the corner wall portions CW1 and CW2 is larger than the plate thickness **t2** of the horizontal wall **30** and the vertical wall **32** and **33** which are the adjacent linear wall portions. In this way, the strength of the corner wall portions CW1 and CW2 of the standing wall **22** is improved. As a result, the electronic apparatus **10** can suppress the corner wall portions CW1 and CW2 being deformed due to an impact from being dropped or the like, and can suppress the glass plate **19** from being broken or the like.

[0074] Here, the result of the drop experiment of the electronic apparatus **10** will be described. In this drop experiment, the electronic apparatus **10** (example) having the recessed hole **40**, the inclined portion **42**, and the thick-walled portion **44** in the corner wall portions CW1 and CW2 and the electronic apparatus (comparative example) not having the recessed hole **40**, the inclined portion **42**, and the thick-walled portion **44** were used. In the drop experiment, the electronic apparatus **10** of the example and the electronic apparatus of the comparative example were freely dropped on the floor from heights of 650 mm, 700 mm, and 750 mm, respectively.

[0075] As a result of the experiment, in the electronic apparatus of the comparative example, the glass plate **19** was not broken at 650 mm, but the glass plate **19** was broken at 700 mm and 750 mm. On the other hand, in the electronic apparatus **10** of the example, the glass plate **19** was not broken at any of 650 mm, 700 mm, and 750 mm. As a result, it was confirmed that the recessed hole **40**, the inclined portion **42**, and the thick-walled portion **44** are effective in suppressing the glass plate **19** from being broken when the electronic apparatus **10** is dropped.

[0076] The present invention is not limited to the above-described embodiment, and it goes without saying that the present invention can be freely modified without departing from the gist of the present invention.

Claims

1. An electronic apparatus comprising: a chassis member having a standing wall on an edge portion of the chassis member; a display panel having a display surface and a rear surface opposite to the display surface and that is supported by an inner surface of the chassis member; and a glass plate configured to cover the display surface of the display panel, wherein the glass plate includes an edge surface of an outer edge portion protruding from an outer peripheral side surface of the display panel and faces the standing wall, wherein an outer edge portion of the glass plate has a corner portion, the standing wall has a corner wall portion that is disposed around the corner portion and has an interior wall surface facing the corner portion, and the interior wall surface of the corner wall portion has a recessed hole recessed in a direction away from the corner portion.
2. The electronic apparatus according to claim 1, wherein an upper end of the recessed hole is at a height equal to or lower than a front surface of the glass plate.
3. The electronic apparatus according to claim 2, wherein a lower end of the recessed hole is lower than a back surface of the glass plate.
4. The electronic apparatus according to claim 1, wherein the interior wall surface of the corner

wall portion has an inclined portion that is inclined to a display panel side toward a base side of the corner wall portion.

5. The electronic apparatus according to claim 1, wherein the standing wall has a linear wall portion that extends along the outer peripheral side surface of the display panel, and a plate thickness of the corner wall portion is larger than a plate thickness of the linear wall portion adjacent to the corner wall portion.

6. The electronic apparatus according to claim 5, wherein the corner wall portion has an exterior wall surface that forms an outer surface of the chassis member on a side opposite to the interior wall surface, the interior wall surface and the exterior wall surface of the corner wall portion have an arc shape, and the plate thickness of the corner wall portion gradually increases towards a center of its corner.

7. The electronic apparatus according to claim 1, further comprising: a first chassis having the chassis member and having a rectangular outer shape; a second chassis having a rectangular outer shape and adjacent to the first chassis; and a hinge configured to rotatably connect one edge portion of the first chassis and one edge portion of the second chassis, wherein the standing wall has: a pair of vertical walls that extend along a pair of edge portions perpendicular to the edge portion of the first chassis to which the hinge is connected and forms the corner wall portion, and a horizontal wall that extends perpendicular to the pair of vertical walls on a side opposite to the edge portion of the first chassis to which the hinge is connected and forms another corner wall portion, and the recessed hole is in each of the corner wall portions between the pair of vertical walls and the horizontal wall.

8. A chassis member for supporting a display assembly including a display panel having a display surface and a glass plate that covers the display surface and is provided such that an outer edge portion of the glass plate protrudes from an outer peripheral side surface of the display panel, the chassis member comprising: a plate portion configured to support a rear surface of the display panel on a side opposite to the display surface; and a standing wall configured to stand upright from an edge portion of the plate portion, wherein the standing wall has a corner wall portion with a recessed hole on an interior wall surface of the standing wall.

9. The chassis member according to claim 8, wherein an interior wall surface of the corner wall portion has an inclined portion that is inclined toward a base side of the corner wall portion.

10. The chassis member according to claim 8, wherein the standing wall has a pair of linear wall portions that are continuous from both ends of the corner wall portions and extend in directions perpendicular to each other, and a plate thickness of the corner wall portion is larger than a plate thickness of the linear wall portions adjacent to the corner wall portion.
